

DATA ANALYSIS INTERNSHIP

Task 10

Movie Analytics Dashboard

What Drives the Success of a Film?
A Power BI Story on 3,754 TMDB Movies

Data Model • KPIs • Genres and Ratings • Box Office • Runtime • Trends • Interactive
Dashboard

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1 Executive summary

The brief asks for an interactive Power BI dashboard that analyses movie data and tells a clear story about the trends and factors related to movie success. This report presents the analysis behind each visual, the data model and the design of the dashboard; the notebook `Movie-Analytics.ipynb` reproduces every number and figure, and the folder `powerbi/` contains the dashboard (`Movie_Analytics_Dashboard.pbix`, with the data loaded), its template (`Movie_Analytics_Dashboard.pbix`), the theme, the DAX measures and the build guide. In the dashboard, the supplied file is loaded as it was received and transformed step by step in Power Query, so the whole path from `tmdb_5000_credits.csv` to every visual can be inspected in Power BI.

Data. The supplied file, `tmdb_5000_credits.csv`, is the primary source: it fixes the list of films and provides their titles, cast and crew (4,803 films). It contains none of the attributes the brief asks to analyse (year, genres, budget, revenue, rating, votes, runtime). These come from a second, public source, TMDb Movies, joined on the TMDb identifier: **3,754 films (78.2%)** of the supplied file, released between 1960 and 2015, are analysed; the unmatched films are mostly 2016 releases.

Key figures. The 3,754 films earned **\$369.5 billion** for **\$130.0 billion** of declared budgets. Among the 2,805 films with both values known, the median film returns **2.27 times** its budget and **76.6%** recover it. The average rating is **6.13 out of 10** and the average runtime **109 minutes**.

Main findings.

- **Money buys revenue, not acclaim.** Budget and revenue are strongly related (Spearman $\rho = 0.65$); budget and rating are not ($\rho = -0.08$).
- **Genre shapes the rating:** Documentary (6.73), War (6.63) and History (6.55) are the best rated, Horror (5.66) the lowest.
- **Longer films are better rated and earn more:** from 5.69 under 90 minutes to 6.93 above 150 minutes, and median revenue triples.
- **The box office is ruled by recent franchise blockbusters** (Avatar, The Avengers, Jurassic World); the best films, once the vote count is considered, are classics with thousands of votes (The Shawshank Redemption, The Dark Knight, The Godfather).
- **The typical film is not richer than thirty years ago** in inflation-adjusted dollars, and average ratings drift down slightly over time.

2 Data and method

2.1 Two sources: the supplied file and a required enrichment

Source 1, supplied: `tmdb_5000_credits.csv`. This is the file received for the task. It is one of the two files of the Kaggle *TMDb 5000 Movie Dataset* (reference 1), hence its name. It defines the population of the study (4,803 films, identified by `movie_id`) and is the only source of the titles, actors and directors used in the dashboard.

Source 2, enrichment: `tmdb-movies.csv`. The brief asks to analyse release years, genres, ratings, budgets, revenues and runtimes. Table 1 shows that the supplied file contains none of them: without a second source, none of the requirements R1 to R8 could be answered, only rankings of actors and directors by number of films. The public TMDb Movies dataset (reference 2) comes from the same database (TMDb) and uses the same identifier, so the two files can be joined without any fuzzy matching.

Table 1: Which source answers each requirement of the brief.

Attribute needed by the brief	Source 1 (supplied)	Source 2 (enrichment)
Film identifier and title	yes	yes
Actors, directors (People page)	yes	no
Release year and date (R1, R7)	no	yes
Genres (R2, R3)	no	yes
Budget and revenue, nominal and 2010 \$ (R4, R5, R7)	no	yes
Rating and vote count (R3, R6, R7)	no	yes
Runtime (R8)	no	yes

Table 2: The two data sources and the join.

File	Content	Films
Source 1: <code>tmdb_5000_credits.csv</code> (supplied)	cast and crew in JSON: actors, billing order, directors	4,803
Source 2: <code>tmdb-movies.csv</code> (public enrichment)	year, genres, budget, revenue (nominal and 2010 \$), rating, votes, runtime	10,866
Inner join on <code>movie_id = id</code>	films of Source 1 found in Source 2	3,754

The join is reliable: 98.3% of the matched films have identical titles, the others carrying their original-language title in the enrichment. The dashboard uses the titles of the supplied file. The companion file of the supplied one, `tmdb_5000_movies.csv` (same Kaggle dataset), would provide the same attributes for all 4,803 films including 2016; it was not supplied, and the Power Query pipeline is written so that it can replace Source 2 by changing a single step.

2.2 Cleaning decisions

- **Zero means unknown.** 564 budgets, 828 revenues and 2 runtimes equal to 0 become missing values. Financial indicators use the 2,805 films with both budget and revenue known.
- **Inflation.** Comparisons across years use the inflation-adjusted revenues (2010 dollars).
- **Weighted rating.** 22.6% of the films have fewer than 50 votes. Films are ranked with the Bayesian average used by IMDb,

$$WR = \frac{v}{v + m} R + \frac{m}{v + m} C,$$

where R is the film's rating, v its votes, $C = 6.13$ the mean rating and $m = 1,354$ the 90th percentile of the vote counts.

- **Multi-valued fields.** Genres (2.65 per film), the five first-billed actors and the directors are unpacked into bridge tables.

2.3 From the supplied file to the Power BI model

The whole preparation runs inside Power BI, in Power Query (Figure 1). The two CSV files are read from a folder set by the parameter `DataFolder`; every step is visible in *Transform data*, in the Applied Steps of each query, so a reader can check how each number of the dashboard is obtained from the supplied file. The notebook performs the same steps in Python and gives the same results.

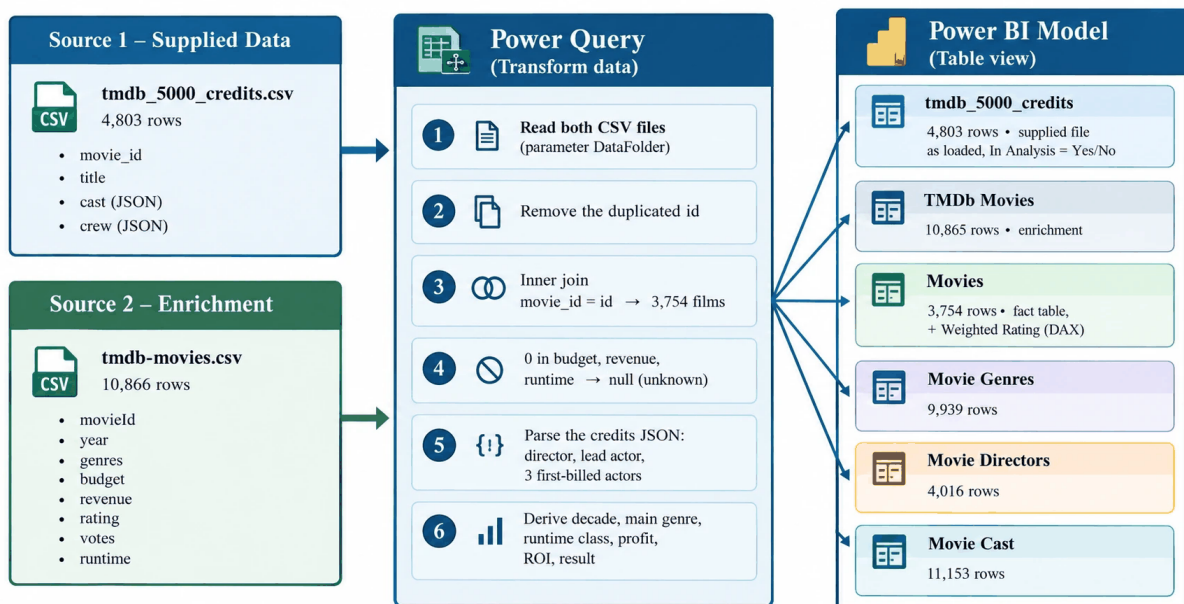


Figure 1: Data pipeline of the dashboard: the supplied file (Source 1) is enriched by TMDb Movies (Source 2) in Power Query, which produces the six tables of the model.

Table 3: Tables of the Power BI model, as listed in the Table view.

Table	Rows	Content
tmdb_5000_credits	4,803	the supplied file as received: <code>movie_id</code> , title, number of cast and crew members, and <i>In Analysis</i> (Yes for the 3,754 films found in Source 2)
TMDb Movies	10,865	the enrichment, after removal of one duplicated identifier
Movies	3,754	fact table, one row per analysed film: year, date, decade, genres, director, lead actor, financials, profit, ROI, result, rating, votes, runtime class, weighted rating
Movie Genres	9,939	one row per film and genre
Movie Directors	4,016	director(s) of each film, from the crew of Source 1
Movie Cast	11,153	three first-billed actors of each film, from the cast of Source 1

Why the analysis tables are separate from the supplied file. The cast and crew columns of `tmdb_5000_credits` hold one JSON text per film, with up to several hundred people inside; Power BI cannot count, filter or rank what is inside such a text. Power Query therefore unpacks them into `Movie Directors` and `Movie Cast`, one row per person and film, and keeps `tmdb_5000_credits` in the model as the reference of what was received. The page *Data Pipeline* of the dashboard (Figure 17) shows this table, its 4,803 films and the share kept in the analysis.

Relationships. `Movies[Movie ID]` is linked one-to-many to the three bridge tables, with a cross-filter direction set to *Both*, so that a genre, actor or director selection filters the films. `tmdb_5000_credits` and `TMDb Movies` are linked to `Movies` on the identifier, which documents the lineage in the Model view.

3 Key performance indicators

Table 4: The eight KPI cards of the Home page (no filter applied).

KPI	Value	Base
Movies	3,754	all matched films
Total revenue	\$369.5 bn	films with known revenue
Total budget	\$130.0 bn	films with known budget
Median ROI (revenue / budget)	2.27x	films with known budget and revenue (2,805)
Profitable films	76.6%	films with known budget and revenue
Average rating	6.13 / 10	all films
Average runtime	109 min	films with known runtime
Total votes	1,912,799	all films

Films with known financials are mostly the larger, better-documented productions; the profitability figures describe the commercial cinema of the dataset rather than every film produced. These eight cards open the Home page of the dashboard (Figure 11).

4 Results

4.1 Movies released over the years (R1)

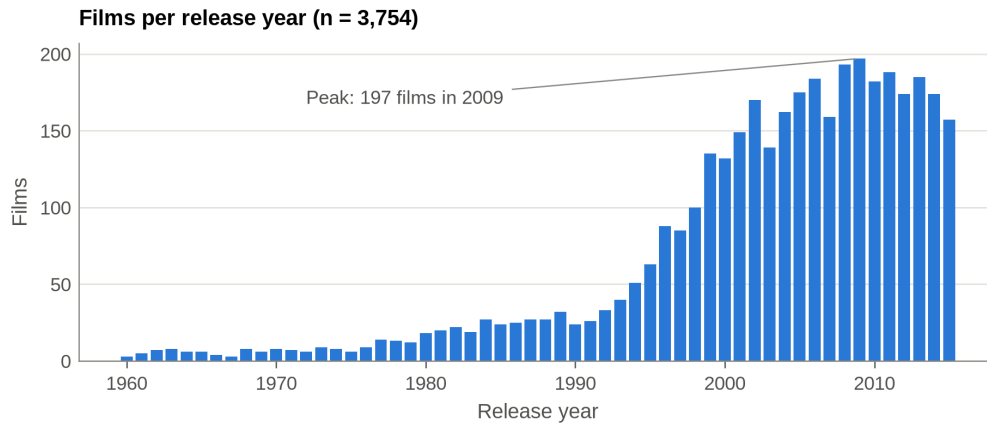


Figure 2: Films per release year. Production rises from a few films a year in the 1960s to 150–200 after 2000, with a peak of 197 in 2009.

72% of the films were released after 2000. The rise reflects both the growth of production and the better documentation of recent films by TMDb users; decades must therefore be compared with averages or medians, never totals. In the dashboard, this chart opens the page *Releases & Genres* (Figure 12).

4.2 Movies by genre (R2)

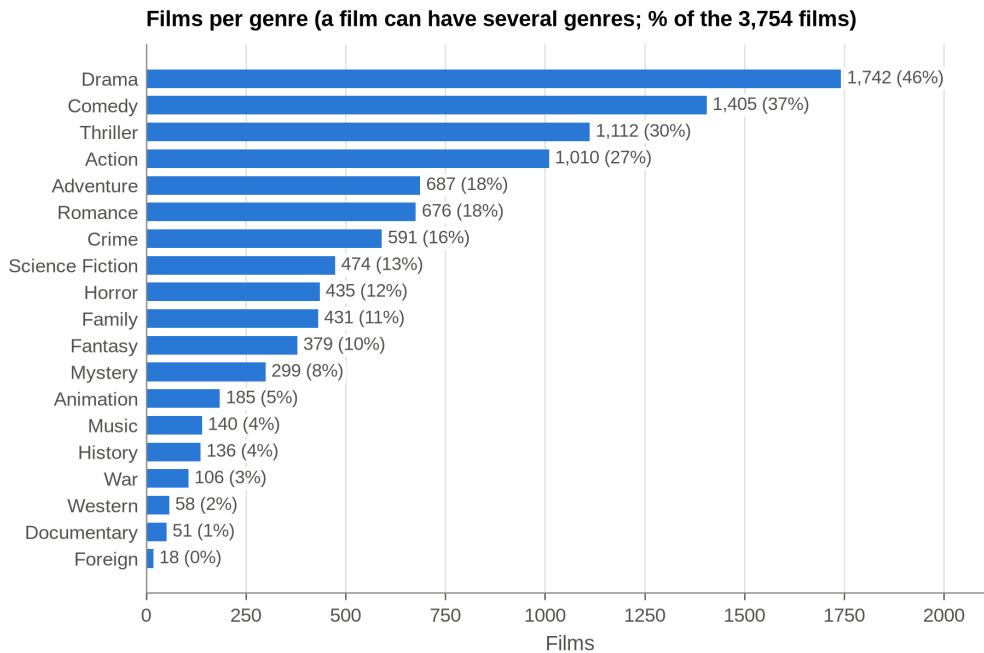


Figure 3: Films per genre. Drama, Comedy, Thriller and Action dominate; a film has 2.65 genres on average, so the bars add up to more than 3,754.

4.3 Average ratings across genres (R3)

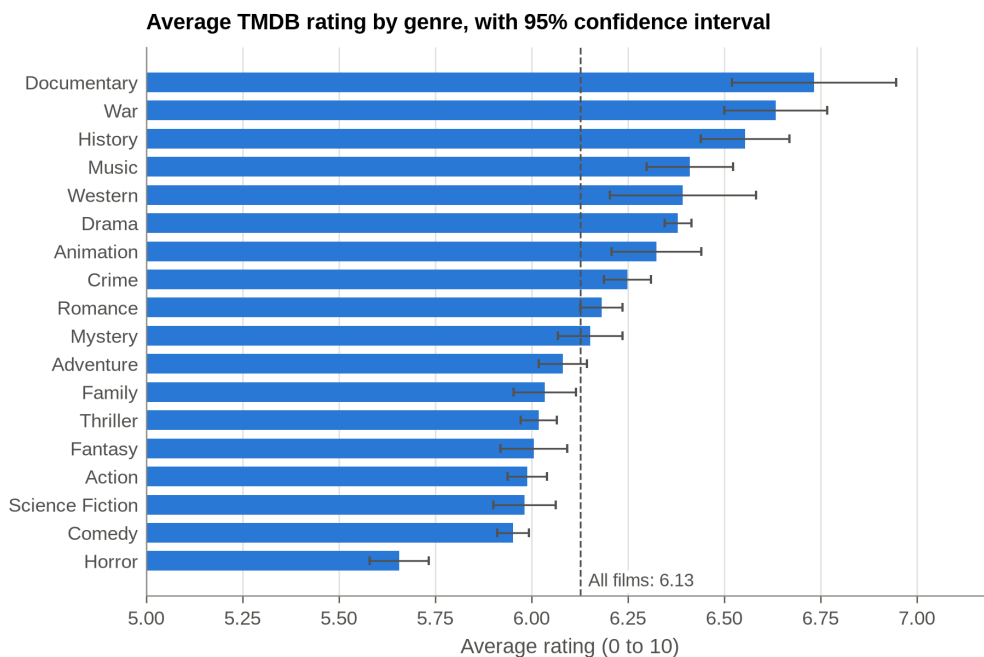


Figure 4: Average rating by genre (genres with at least 50 films), with 95% confidence intervals. The dashed line is the average of all films (6.13).

Documentary, War, History, Music, Western and Drama are rated above average; Horror is clearly the lowest (5.66), followed by Comedy, Science Fiction and Action (about 6.0). The genre effect is significant (Kruskal–Wallis $H = 626$, $p < 10^{-100}$) but spans only about one point on the ten-point scale. The dashboard shows the same ranking on the page *Releases & Genres* (Figure 12, lower left).

4.4 Budget and revenue (R4)

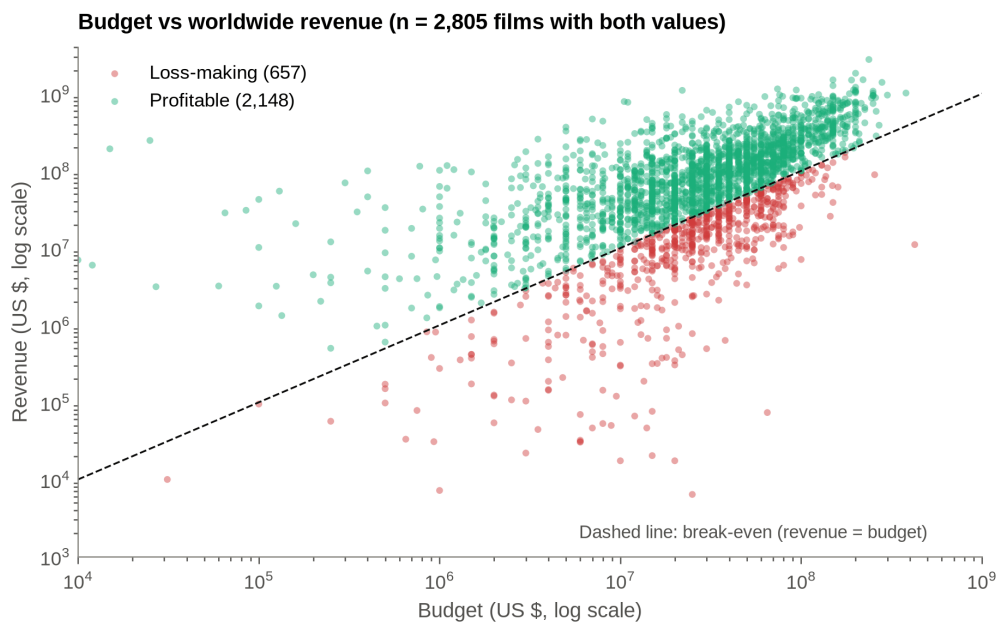


Figure 5: Budget against worldwide revenue, logarithmic scales. Films above the dashed line recover their budget (green), films below do not (red).

Table 5: Return on investment by budget class (films with known financials).

Budget class	Films	Median revenue (\$M)	Median ROI	Profitable
< \$10M	568	13.9	3.39	76%
\$10–30M	926	41.8	2.17	74%
\$30–70M	793	95.0	1.98	75%
\$70–150M	424	237.5	2.29	82%
> \$150M	94	554.9	2.85	95%

Bigger budgets bring bigger revenues ($\rho = 0.65$) and fewer failures, but not a better return: the median ROI is highest for small films (3.4x below \$10M). The page *Box Office* of the dashboard (Figure 13) shows the same scatter for each film, with the median ROI by main genre: Horror (3.35x) and Science Fiction (3.10x) lead.

4.5 Highest-grossing movies (R5)

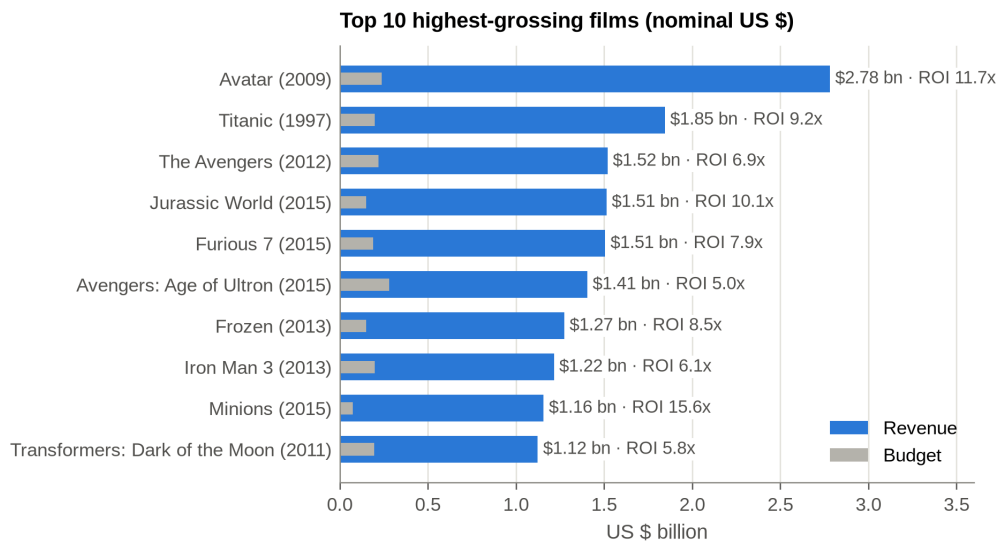


Figure 6: Top 10 films by worldwide revenue (nominal dollars), with their budget (grey bar) and return on investment.

Avatar (\$2.78 bn), Titanic (\$1.85 bn) and The Avengers (\$1.52 bn) lead. All are franchise or animation blockbusters released after 2009, except Titanic, and each returns 5 to 16 times its budget (Figure 13, top right). In 2010 dollars, older hits enter the top 10: Star Wars (1977), The Exorcist (1973), Jaws (1975) and E.T. (1982).

4.6 Highest-rated movies considering the vote count (R6)

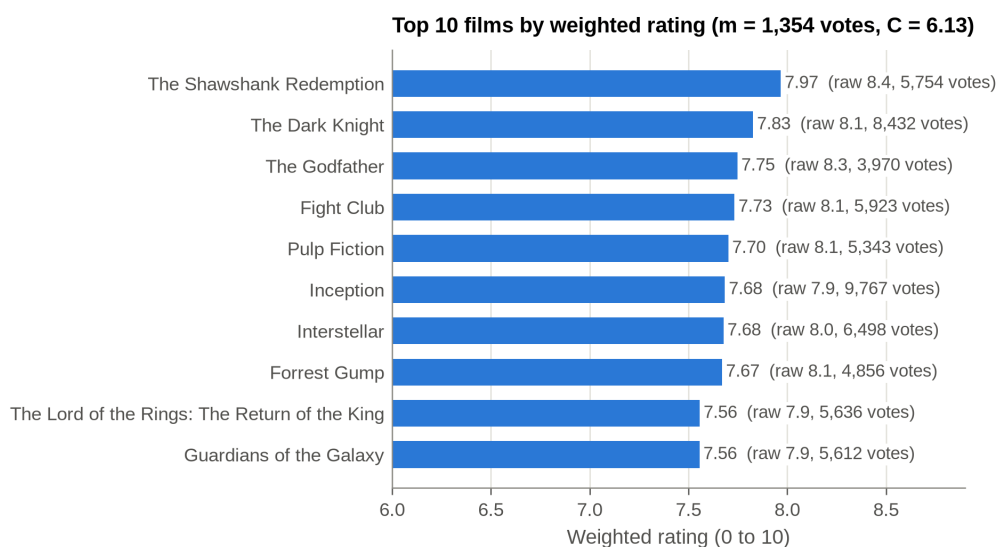


Figure 7: Top 10 films by weighted rating. The raw rating and the number of votes are given for each film.

On the raw rating, a film rated 8.4 by 11 users ties with *The Shawshank Redemption* (8.4 from 5,754 users). The weighted rating removes this artefact: the top 10 are all films with thousands of votes. The same ranking is the table of the page *Ratings & Runtime* (Figure 14), sorted by weighted rating.

4.7 Revenue and ratings over the years (R7)

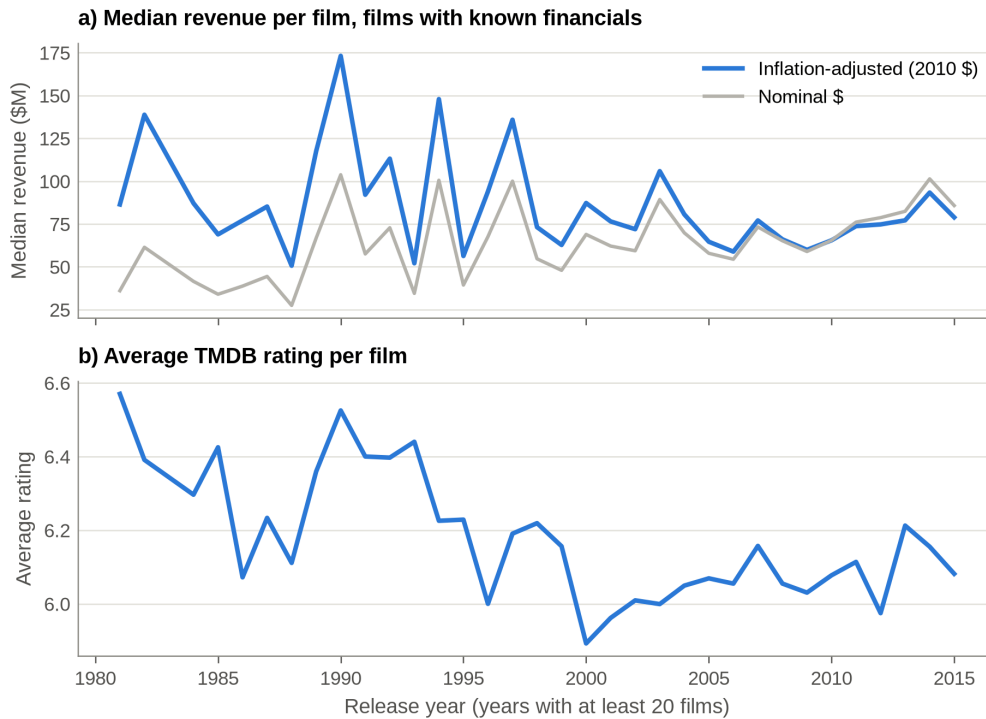


Figure 8: (a) Median revenue per film, inflation-adjusted and nominal; (b) average rating per film. Years with at least 20 films.

In 2010 dollars, the median film falls from about \$136M in the 1970s to \$81–84M in the 1980s–1990s and \$67–68M after 2000; nominal growth is mostly inflation. The average rating drifts from about 6.6 for films of the 1960s–1970s to 6.0–6.1 after 2000 ($\rho = -0.09$), partly because the old films kept in the database are the classics that people still watch and rate.

4.8 Runtime, ratings and revenue (R8)

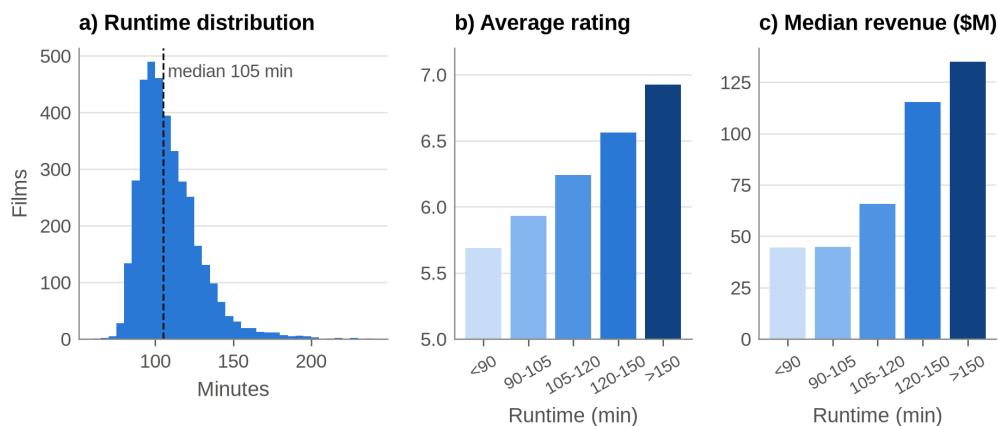


Figure 9: (a) Runtime distribution (median 105 minutes); (b) average rating and (c) median revenue by runtime class.

Ratings rise steadily with length, from 5.69 below 90 minutes to 6.93 above 150 minutes ($\rho = 0.38$), and the median revenue triples from about \$45M to \$135M ($\rho = 0.25$). This is an association: long films are more often prestige dramas, epics and blockbusters; length alone does not make a film better. The two runtime charts of Figure 14 exclude the two films of unknown runtime.

4.9 Complement: directors and actors (from the supplied credits)

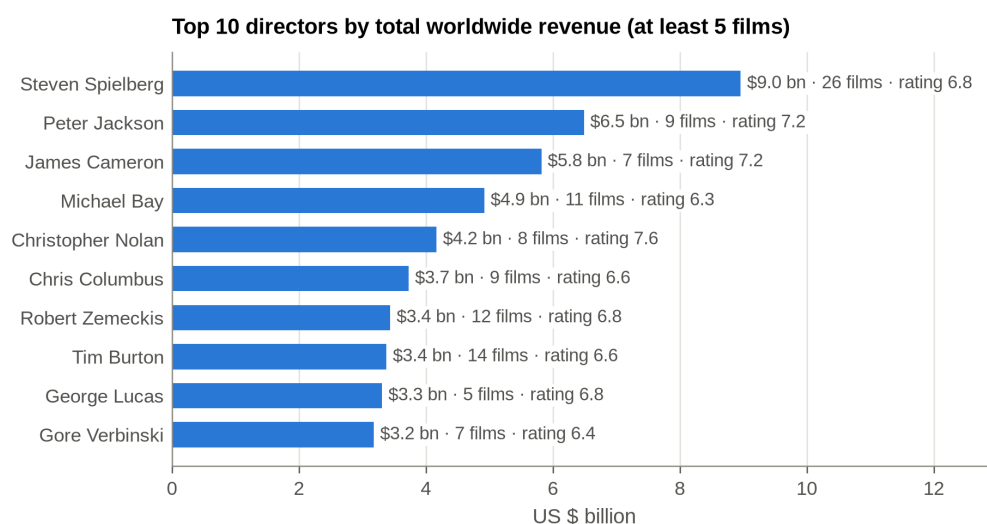


Figure 10: Top 10 directors by total worldwide revenue (at least five films with known financials).

Steven Spielberg (\$9.0 bn, 26 films), Peter Jackson and James Cameron lead; Christopher Nolan combines box office with the best average rating (7.64). The most frequent actors in the first five billings are Robert De Niro (51 films), Samuel L. Jackson (41) and Bruce Willis (39). The

dashboard counts the three first-billed actors only, the leading roles: Robert De Niro (44), Bruce Willis (35), Matt Damon and Nicolas Cage (34). Its ranking of the best-rated directors uses the weighted rating and directors with at least three films, so that one acclaimed film is not enough: Christopher Nolan (7.28), Pete Docter (7.25) and Quentin Tarantino (6.95) lead (Figure 15).

4.10 Statistical tests

Table 6: Spearman rank correlations between success factors.

Pair	n	ρ	Strength
Budget – revenue	2,805	0.65	strong
Vote count – rating	3,754	0.39	moderate
Runtime – rating	3,752	0.38	moderate
Runtime – revenue	2,805	0.25	weak
Rating – revenue	2,805	0.19	weak
Release year – rating	3,754	−0.09	negligible
Budget – rating	2,805	−0.08	negligible

With nearly 3,000 films, all these correlations are statistically significant; the strength, not the p-value, says what matters.

5 The Power BI dashboard

The dashboard is delivered in two forms, in the folder `powerbi/`:

- `Movie_Analytics_Dashboard.pbix`: the dashboard with the data loaded. It opens directly in Power BI Desktop, with the seven pages shown in the captures below.
- `Movie_Analytics_Dashboard.pbix`: the same dashboard as a template, without data. On opening, Power BI asks for `DataFolder`, the folder that holds `tmdb_5000_credits.csv` and `tmdb-movies.csv`, and rebuilds every table from these two files.

Both contain the Power Query pipeline, the data model (six tables, five relationships, twenty-two DAX measures and one calculated column), the theme and seven pages. Its visual style follows the OTT reference dashboard: dark navy background, light-blue accents, a glowing header band with a page navigator, rounded frames behind every visual, and two custom visuals (word cloud and scrolling text).

Table 7: Pages of the dashboard and the requirements they answer.

Page	Visuals
1. Home	eight KPI cards (R9); word cloud of genres; three clickable tiles opening the analysis pages; scrolling list of the highest-grossing titles
2. Releases & Genres	films per year (R1); films by genre (R2); average rating by genre and by decade (R3)
3. Box Office	budget vs revenue scatter with trend line (R4); top 10 highest-grossing films (R5); median inflation-adjusted revenue per year, years with at least 20 films (R7); median ROI by main genre
4. Ratings & Runtime	top 10 films by weighted rating (R6); gauge of the average rating; rating and median revenue by runtime class (R8); average rating over the years (R7)
5. People	top 10 directors by revenue; top 10 actors by number of films; best-rated directors (weighted score, at least three films)
6. Insights	the ten insights of Section 6 (R11)
7. Data Pipeline	the supplied file as loaded (4,803 films), films kept in the analysis (3,754, 78.2%), cast and crew entries, and the six steps of Figure 1

Interactivity (R10). Four slicers synchronised across the analysis pages (release year, genre, runtime class, financial result); cross-filtering between all visuals (clicking a bar filters the page); Top N filters that react to the slicers; a page navigator in the header and clickable tiles on the Home page; tooltips on every visual. The DAX measures and the build steps are in the folder `powerbi/`: `DAX_measures.dax` and `Dashboard_Build_Guide.md`.

The figures below are captures of the seven pages of the dashboard, with no filter applied; every value they show matches the notebook and Sections 3 and 4.

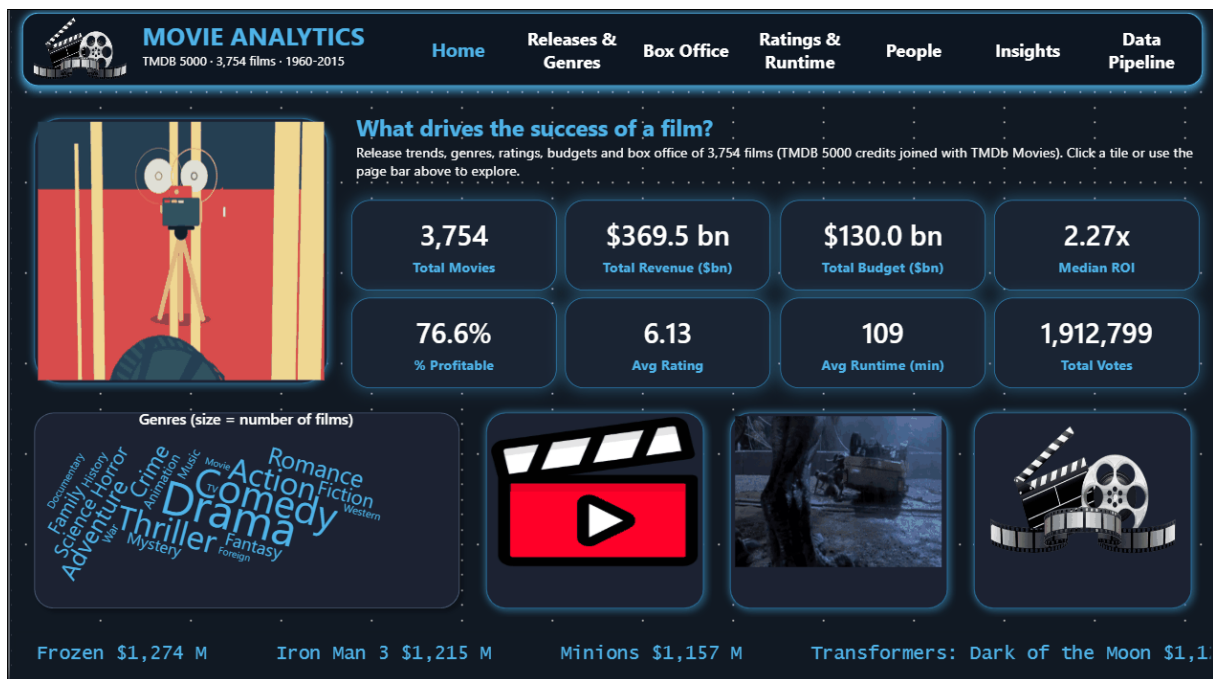


Figure 11: Home page: eight KPI cards on the whole dataset (3,754 films, \$369.5 bn of revenue, \$130.0 bn of budget, median ROI 2.27x, 76.6% profitable, average rating 6.13, average runtime 109 minutes, 1,912,799 votes), word cloud of genres, navigation tiles and the scrolling list of top-grossing titles (Avatar \$2,782M, Titanic \$1,845M).



Figure 12: Releases & Genres: production rises after the mid-1990s and peaks at 197 films in 2009; Drama (1,742 films), Comedy (1,405), Thriller (1,112) and Action (1,010) dominate; Documentary (6.73), War (6.63) and History (6.55) are the best-rated genres; the average rating declines from 6.66 (1970s) to 6.03 (2000s).



Figure 13: Box Office: revenue grows with budget (trend line); Avatar, Titanic and The Avengers lead the box office; the median inflation-adjusted revenue per film (years with at least 20 films, from 1981) is lower after 2005 than in the 1980s and 1990s; Horror (3.35x) and Science Fiction (3.10x) have the highest median return on investment.



Figure 14: Ratings & Runtime: the ten best films once the vote count is taken into account, led by The Shawshank Redemption (7.97), The Dark Knight (7.83) and The Godfather (7.75); the average rating of 6.13 out of 10; longer films are better rated (5.69 to 6.93) and earn a higher median revenue (\$39M to \$133M).

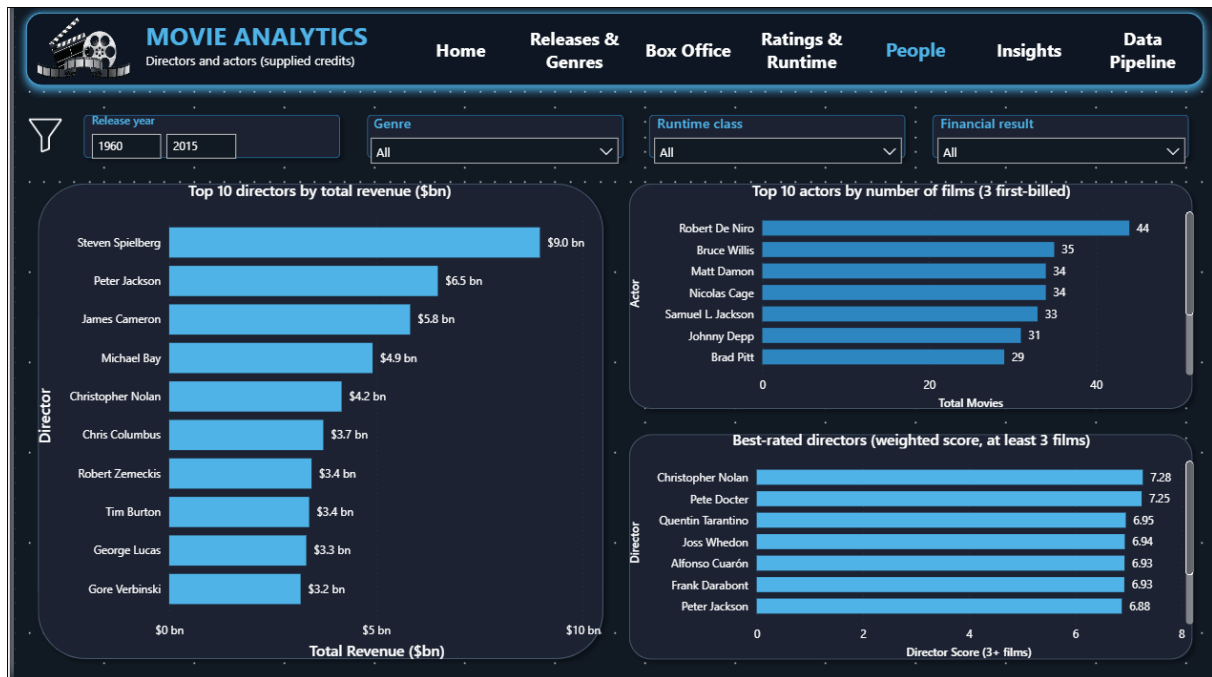


Figure 15: People: Steven Spielberg (\$9.0 bn), Peter Jackson and James Cameron lead the directors by revenue; Robert De Niro (44 films), Bruce Willis and Matt Damon are the most frequent first-billed actors; Christopher Nolan (7.28) and Pete Docter (7.25) have the best weighted score among directors with at least three films.

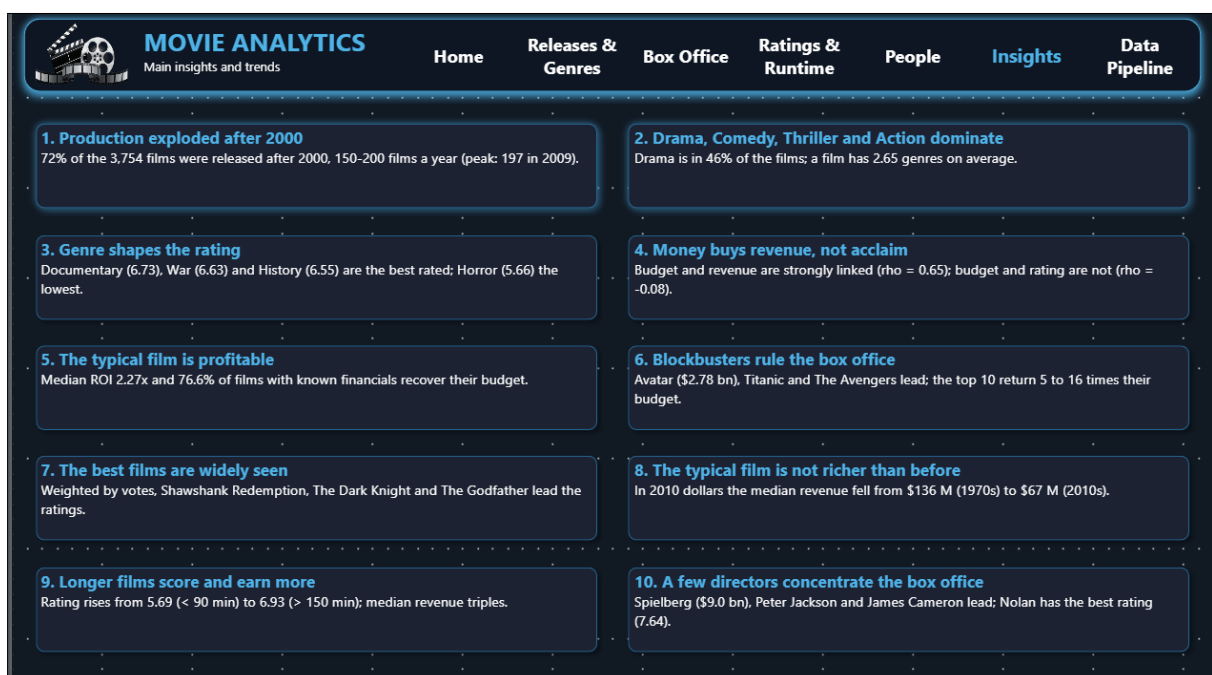


Figure 16: Insights: the ten insights of Section 6, as text cards that summarise the story of the dashboard.

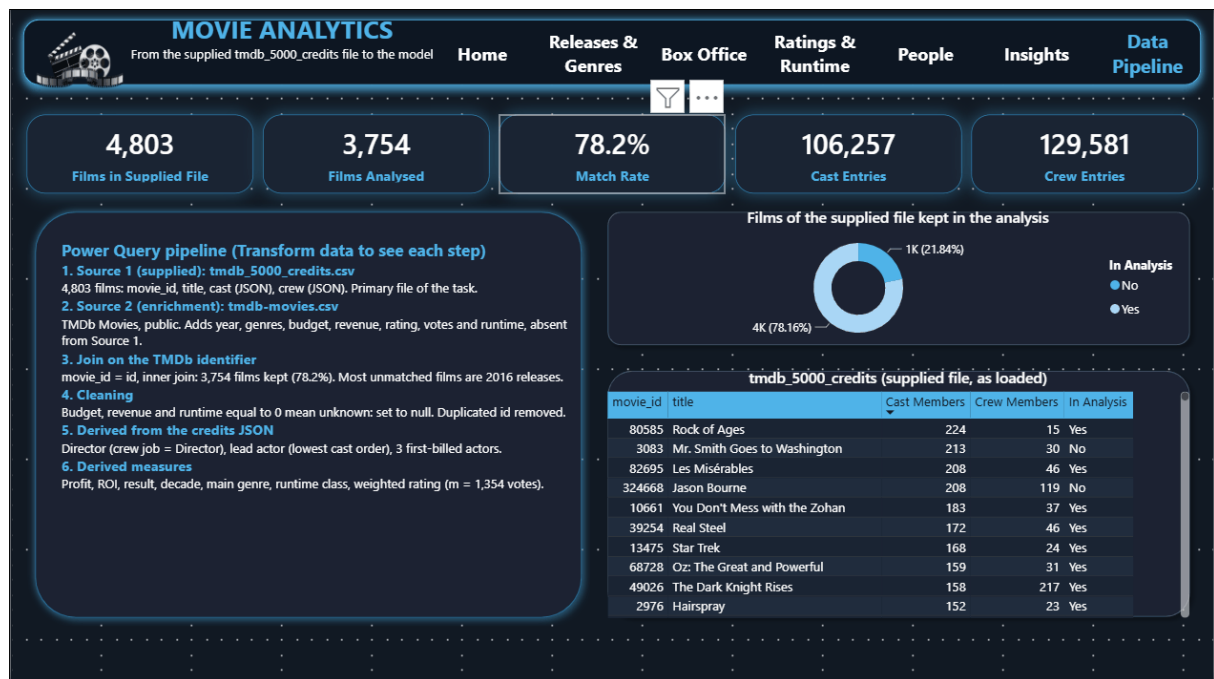


Figure 17: Data Pipeline: the supplied file `tmdb_5000_credits` as loaded in Power BI (4,803 films, 106,257 cast and 129,581 crew entries), the 3,754 films kept in the analysis (78.2%; 1,049 films not found in Source 2), the table of the supplied file with its *In Analysis* flag, and the six steps of the Power Query pipeline.

6 Insights and trends

These ten insights form the page *Insights* of the dashboard (Figure 16).

- Production exploded after 2000:** 72% of the films, 150–200 a year.
- Drama, Comedy, Thriller and Action dominate** the catalogue.
- Genre shapes the rating:** Documentary, War and History about half a point above average, Horror half a point below.
- Money buys revenue, not acclaim:** budget is strongly linked to revenue, not to rating.
- The typical film is profitable:** median ROI 2.27x, 76.6% recover their budget, but one in four does not.
- The box office is ruled by recent franchise blockbusters;** in adjusted dollars, classics still rival them.
- The best films are also widely seen:** the weighted top 10 are films with thousands of votes.
- The typical film is not richer than thirty years ago** in adjusted dollars, and ratings drift down slightly.
- Longer films are better rated and earn more,** an association driven by the type of film.
- A few directors concentrate the box office;** Nolan combines revenue and the best average rating.

Recommendations for a studio. Use budget as a revenue lever, not a quality lever; build

critical reputation in Drama, History and War while keeping profitable genres such as Horror and Comedy in the slate; judge films on ROI and adjusted dollars; rank quality with the weighted rating and always show the vote count.

7 Limitations

- 3,754 of the 4,803 supplied films are analysed; 2016–2017 releases are missing because the enrichment stops in 2015. With the companion file `tmdb_5000_movies.csv`, only the loading cell of the notebook and the query `TMDb Movies Raw` of the dashboard need to change.
- Budgets and revenues are declared figures, known for 2,805 films; small productions are under-represented.
- TMDb ratings come from users, not critics, and over-represent popular films.
- The correlations are associations, not causes.

8 Reproducibility

Every number, table and figure of this report is produced by the notebook `Movie-Analytics.ipynb` from the two files of `data/raw/`: the supplied `tmdb_5000_credits.csv` and the enrichment `tmdb-movies.csv`. The notebook writes the analysis tables (`data/powerbi/`), the result tables (`results/`) and the figures (`figures/`). The dashboard `powerbi/Movie_Analytics_Dashboard.pbix` shows the result; its template `powerbi/Movie_Analytics_Dashboard.pbix` rebuilds the same tables in Power Query from the same two files; opening it with the folder `data/raw/` as `DataFolder` reproduces every page shown in Section 5. The whole project is published at <https://github.com/koffifidelek59-collab/movie-analytics>.

References

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